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NATIONAL UNIVERSITY OF TECHNOLOGY

(Department of Computer Science)

CS161– Database Systems

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LAB TASK 12

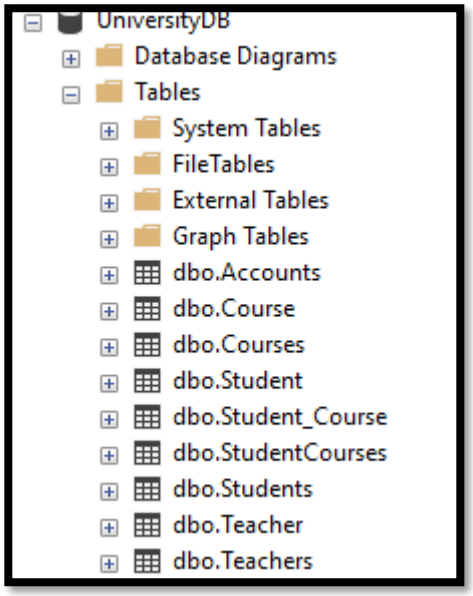
In Lab Tasks

Task 1

Normalize the given table up to 3NF.

STU_ID	COR_ID	STU_Name	CORE_ name	Grade	Faculty	Teacher_ Phone
1	CS012	Faqeer	Database	A	Hakeem	111
1	CS018	Faqeer	Program	B	Yaqoob	150
2	CS012	Jalil	Database	A	Hakeem	111
3	CS012	Sameer	Database	B	Hakeem	111
4	CS018	Bhola	Program	A	Yaqoob	150
4	CS012	Bhola	Database	B	Hakeem	111

Create the normalized database in Microsoft SQL Server at your workstations.



```sql
CREATE TABLE Teacher (
    Faculty VARCHAR(50) PRIMARY KEY,
    Teacher_Phone INT
);

CREATE TABLE Student_Course (
    STU_ID INT,
    COR_ID VARCHAR(10),
    Grade CHAR(1),
    PRIMARY KEY (STU_ID, COR_ID),
    FOREIGN KEY (STU_ID) REFERENCES Student(STU_ID),
    FOREIGN KEY (COR_ID) REFERENCES Course(COR_ID)
);

ALTER TABLE Course
ADD FOREIGN KEY (Faculty) REFERENCES Teacher(Faculty);
```

Task 2: Transactions

Table Structure:

```sql
CREATE TABLE Accounts (
 AccountID INT PRIMARY KEY,
 AccountHolder VARCHAR(100),
 Balance DECIMAL(10, 2)
);
```

Sample Data:

```sql
INSERT INTO Accounts (AccountID, AccountHolder, Balance)
VALUES
 (1, 'Alice Johnson', 5000.00),
 (2, 'Bob Smith', 3000.00),
 (3, 'Charlie Davis', 7000.00);
```

Results | Messages

	AccountID	AccountHolder	Balance
1	1	Alice Johnson	5000.00
2	2	Bob Smith	3000.00
3	3	Charlie Davis	7000.00

1. Money Transfer (Using Transaction)
   - Start a transaction.
   - Subtract ₹1000 from AccountID 1.
   - Add ₹1000 to AccountID 2.
   - **Commit the transaction.**

```sql
BEGIN TRANSACTION;

UPDATE Accounts
SET Balance = Balance - 1000
WHERE AccountID = 1;

UPDATE Accounts
SET Balance = Balance + 1000
WHERE AccountID = 2;

COMMIT TRANSACTION;
```

144 %

Messages

```
(1 row affected)

(1 row affected)

Completion time: 2025-05-20T13:36:40.2840416+05:00
```

2. **ROLLBACK** Example
   - Start the same transaction.
   - After subtracting ₹1000, cause an error (e.g., invalid column).
   - Use ROLLBACK to undo all changes.

```sql
BEGIN TRANSACTION;
UPDATE Accounts
SET Balance = Balance - 500
WHERE AccountID = 1;

SAVE TRANSACTION SavePoint1;

ROLLBACK TRANSACTION SavePoint1;

COMMIT TRANSACTION;
```

144 %

Messages

```
(1 row affected)

Completion time: 2025-05-20T13:39:47.6077771+05:00
```

3. **SAVEPOINT** Example
   - Start a transaction.
   - Subtract ₹500 from AccountID 1.
   - Set a SAVEPOINT.
   - Subtract another ₹500.
   - Roll back to the SAVEPOINT.
   - Commit the transaction.

```sql
BEGIN TRANSACTION;

UPDATE Accounts
SET Balance = Balance - 500
WHERE AccountID = 1;

SAVE TRANSACTION SavePoint1;

UPDATE Accounts
SET Balance = Balance - 500
WHERE AccountID = 1;

ROLLBACK TRANSACTION SavePoint1;

COMMIT TRANSACTION;
```

144 %

Messages

```
(1 row affected)

(1 row affected)

Completion time: 2025-05-20T13:40:45.7685064+05:00
```

Task 3 Backup

1. Create a Backup Demo

- Create a new database named LabBackupDemo.
- Create a table Students with 5 sample records.

```sql
-- Create Students table
CREATE TABLE Students (
 StudentID INT PRIMARY KEY,
 Name VARCHAR(50),
 Age INT,
 Course VARCHAR(50)
);
```

-- Insert 5 sample records
INSERT INTO Students (StudentID, Name,
Age, Course) VALUES (1, 'Ali Khan', 20,
'Computer Science'),
(2, 'Fatima Malik', 22, 'Information Technology'),
(3, 'Usman Ahmed', 21, 'Software Engineering'),
(4, 'Sara Iqbal', 19, 'Data Science'),
(5, 'Ahmed Raza', 23, 'Artificial Intelligence');

```sql
CREATE DATABASE LabBackupDemo;
GO
USE LabBackupDemo;
GO
CREATE TABLE Students (
    StudentID INT PRIMARY KEY,
    Name VARCHAR(50),
    Age INT,
    Course VARCHAR(50)
);
GO
INSERT INTO Students (StudentID, Name, Age, Course) VALUES
(1, 'Ali Khan', 20, 'Computer Science'),
(2, 'Fatima Malik', 22, 'Information Technology'),
(3, 'Usman Ahmed', 21, 'Software Engineering'),
(4, 'Sara Iqbal', 19, 'Data Science'),
(5, 'Ahmed Raza', 23, 'Artificial Intelligence');
```

144 %

Messages

```
(5 rows affected)

Completion time: 2025-05-20T13:46:42.9035556+05:00
```

2. Full Backup

· Take a full backup of the LabBackupDemo database.

```sql
USE LabBackupDemo;
GO
CREATE TABLE Students (
    StudentID INT PRIMARY KEY,
    Name VARCHAR(50),
    Age INT,
    Course VARCHAR(50)
);
```

```sql
INSERT INTO Students (StudentID, Name, Age, Course) VALUES
(1, 'Ali Khan', 20, 'Computer Science'),
(2, 'Fatima Malik', 22, 'Information Technology'),
(3, 'Usman Ahmed', 21, 'Software Engineering'),
(4, 'Sara Iqbal', 19, 'Data Science'),
(5, 'Ahmed Raza', 23, 'Artificial Intelligence');
```

3. Differential Backup

- Update some student records (e.g., change a name or age).
- Take a differential backup (only the changed data).

```sql
RESTORE DATABASE LabBackupDemo
FROM DISK = 'D:\LabBackupDemo\LabBackupDemo_full.bak'
WITH NORECOVERY;
```

4. Transaction Log Backup

- Perform a few more updates or deletes.
- Take a transaction log backup to capture recent changes.

```sql
USE LabBackupDemo;
UPDATE Students
SET Name = 'Ali Khan', Age = 21
WHERE StudentID = 1;
```

5. Restore from Full Backup

- Drop or delete the LabBackupDemo database.
- Restore it using the full backup only.

```sql
UPDATE Students
SET Age = 22
WHERE StudentID = 2;

DELETE FROM Students
WHERE StudentID = 3;
```

6. Restore to Latest State

· Restore the full backup.

· Apply the differential and transaction log backups to bring back the latest state.

```sql
BACKUP LOG LabBackupDemo
TO DISK = 'D:\LabBackupDemo\LabBackupDemo_full.bak'
```

7. Recover from User Error

· Simulate a user error (e.g., delete some student data).

· Use the log backup to recover the database to a point before the error happened.

```sql
USE master;
ALTER DATABASE LabBackupDemo SET SINGLE_USER WITH ROLLBACK IMMEDIATE;
DROP DATABASE LabBackupDemo;
```

8. Recovery Options

· Try restoring with:

- RESTORE WITH RECOVERY
- RESTORE WITH NORECOVERY

```sql
RESTORE DATABASE LabBackupDemo
FROM DISK = 'D:\LabBackupDemo\LabBackupDemo_full.bak'
WITH NORECOVERY;
```

Post Lab Tasks

Task 1

Report Generation:

- Write a detailed report for the in-lab tasks along with all steps taken to achieve the above task.

Lifelong Learning Component

Reflective Learning Task: In this lab, you practiced creating backups and restoring databases in SQL Server. These skills are critical for protecting data and ensuring business continuity. Reflect on the importance and practical applications of database backups and recovery:

- Why is it essential for organizations to perform regular full, differential, and transaction log backups?

  ANS:
  Organizations rely on regular **full, differential, and transaction log backups** to safeguard their data against hardware failures, cyberattacks, and accidental deletions. These backups ensure that critical information is not lost and can be restored efficiently.
  - **Full Backups:** Capture the entire database, providing a comprehensive restore point.
  - **Differential Backups:** Store changes made since the last full backup, reducing storage needs while speeding up recovery.
  - **Transaction Log Backups:** Record all transactions since the last backup, allowing point-in-time recovery.

- How do different types of backups help in saving time and storage space while ensuring data safety?

  ANS:
  Using a combination of backup types optimizes resources. Instead of performing frequent full backups, organizations can rely on differential and transaction log backups to minimize storage consumption and reduce downtime during restoration.

- Consider a scenario where a user accidentally deletes important data. How does having transaction log backups help restore the database to the exact state before the error?

  ANS:
  If a user accidentally deletes important data, transaction log backups allow the database to be restored to the exact state before the error. This method helps maintain data consistency and ensures minimal disruption to operations**.**

- What are the benefits of using NORECOVERY mode during restore operations, especially when applying multiple backups?

  ANS:
  When applying multiple backups in a restore sequence, **NORECOVERY mode** ensures that the database remains in a non-operational state until the full recovery process is complete. This is useful when applying a sequence of backups without prematurely making the database accessible.

- How might you explain the importance of backup and recovery plans to non-technical stakeholders in a company?

  ANS:
  To non-technical stakeholders, backup and recovery plans can be framed as insurance for the company's data. Without proper backups, business operations can halt, leading to financial loss and reputational damage. Simplifying the explanation helps them understand why investing in backup solutions is crucial.